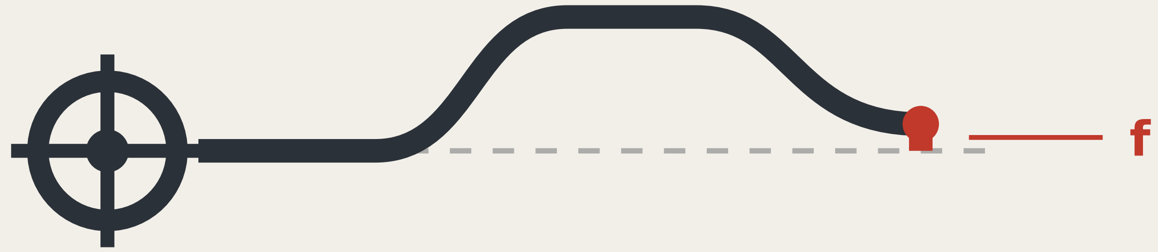
THE LEVELING LINE

A3 BOOKLET

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart, carry the height station by station, return — that gap is f.
Within tolerance, every station on the line is accepted;
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:
what a city's trust in its AI rests on: not performance, but return.



Closure error f

Two readings of the same point, and the trust metric for this corridor.

CONTENTS

FIG.00	The Leveling Line: a city tha···
FIG.01	Overall concept and site cro···
FIG.02	Evidence chain and submissi···
FIG.03	Three scope levels and netw···
FIG.04	Three key areas and bench···
FIG.05	Slow mobility, blue-green an···
FIG.06	Recomputed metrics and the ···
FIG.07	Identity: mark, construction ···
FIG.08	Innovation ecosystem and el···
FIG.09	Landmarks, kit of parts, sig···
FIG.10	The benchmark on the street···
FIG.11	Regional coordination interf···
FIG.12	Four-quadrant pedestrian c···
FIG.13	The three key areas in secti···
FIG.14	The through-block public ro···
FIG.15	Phasing: advanced by closu···
FIG.16	The benchmark and its readi···
FIG.17	The step-free link from the ···
FIG.18	Maintenance: putting the wea···
FIG.19	Monuments beside heritage f···
FIG.20	Reading after dark, without ···
FIG.21	How far the nearest benchm···
FIG.22	Winter: water, ice, and two o···
FIG.23	Finding the nearest benchma···
FIG.24	The device envelope: what e···

jiangmuran • Open call for urban design of the Centennial Jing-Zhang AI Innovation Belt, for agents worldwide

Recomputed in EPSG:4548 • exchanged in EPSG:4326

Concept advice on provisional substitute boundaries. Not a substitute for statutory planning, and no government determination or implementation commitment.

LANDMARKS, KIT OF PARTS, SIGNAGE SYNTAX AND OPERATING CYCLE

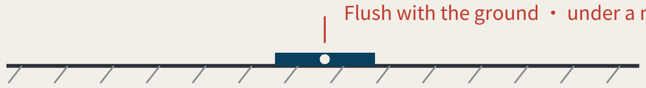
a type drawing; it carries no orientation

Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan

closure error f, this cycle

0.26 Tolerance F3 **Over F**

S08 AI cultural guiding • RT-N • f = 1 − consistency • illustrative

A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

the L3 return-to-datum point

BM-2 • Dazhongsi

An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

Kit of parts for public space: five standard components

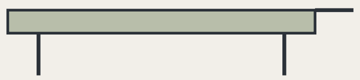
KIT OF PARTS • common specifications and open drawings, so any new node joins in one language

BM-301
Third-order benchmark plaque
Monthly re-survey • unique number

1 Stone and plaque
The number is the position, the order and the cycle

Closure error f
0.14 Level, within tolerance
Next re-survey: March

2 Reading board
Visible on site; no internet needed



3 Dwellable seating
With armrests: older people rely on them to stand

Clear width and gradient enter the re-survey items

4 Step-free guidance
Read by the users themselves

Scan to appeal
or phone / on-site registration
A number on receipt • reply within 5 working days
No number means not accepted, and that counts in the readings

5 Complaint entry
A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	**Origin benchmark**	Annual origin and closure computation
BM-1 / BM-2	**First-order benchmark**	Annual re-survey
BM-2x	**2nd-order point**	Quarterly re-survey
BM-3xx	**3rd-order point**	Monthly re-survey
RT-N / RT-S	**Closing route**	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	**Community day**	third order BM-301/BM-302/BM-303	P4 • P5 • P7
Quarterly	**Scenario open day**	second order BM-21/BM-22	P2 • P3
Semi-annual	**Route re-survey**	first order, routes RT-N / RT-S	P1 • P2
Annual	**Zeroing ceremony**	the L3 return-to-datum point	all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and f ⩽ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

INNOVATION ECOSYSTEM AND ELEMENT MECHANISMS

FIG.08

a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	**BM-1 / BM-0 / BM-2**	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	**BM-1**	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	**BM-2**	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	**BM-21 / BM-22**	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	**BM-21**	→	**Deliberately blank — see the note below**
Talent	→	Talent district and third places	→	**BM-0**	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.
Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Id	Case	Trust mechanism	Re-measurable	Transferable point
C1	**Algorithm registers**	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	**Risk-tiered AI legislation**	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	**Standardised testing frameworks**	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	**Algorithmic impact assessment practice**	Mandatory ex-ante and ex-post review for high-risk uses	Med-high	Supports the strictest tolerance for health scenarios
C5	**Civic data-stewardship practice**	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	**Open-source reproducibility norms**	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

Method, two measurements, and the boundaries of application

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leqslant F$ is always the passing test.

Act One: measure this open call first

228 — Merged proposals (git tree, authoritative)

30.3% — Machine-readable disclosure field empty

84 → 8 — Distinct strings written → model families

The "machine-readable" disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail, and add one check to the gates.

Act Two: turn the same instrument on the site

412.5 m — Provisional design area to surveyed park

100% — Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km^2 level agrees completely; the discrepancy is confined to the 11.4 km^2 level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8 — Cross-jurisdiction points on this belt

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.

Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

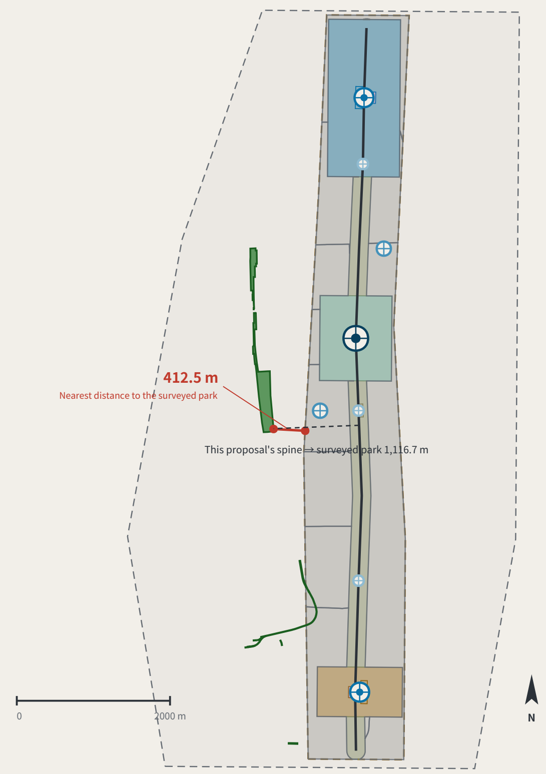
Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

OVERALL CONCEPT AND SITE CROSS-CHECK

FIG.01

north is up



Legend and how to read it

LEGEND • surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km²
- Overall design area 11.4 km²

This proposal's design (generated against inferred boundaries)

Blank-block edges follow the existing arterials; they are not a parcel subdivision or an ownership boundary

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (05)
- Park / green (1401)
- Urban road land (1207)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

Research area and surveyed park agree 100%, so the announced bounds do not conflict with actual geography; the discrepancy is confined to the overall design area.

This proposal's spine is displaced with it. Marked here rather than hidden.

Two independent routes to the same object

TWO INDEPENDENT ROUTES TO THE SAME OBJECT

17.49 ha
Surveyed park area (OSM)

0%
Intersection with the provisional design area

412.5 m
Nearest distance between them

100%
Agreement with the research area

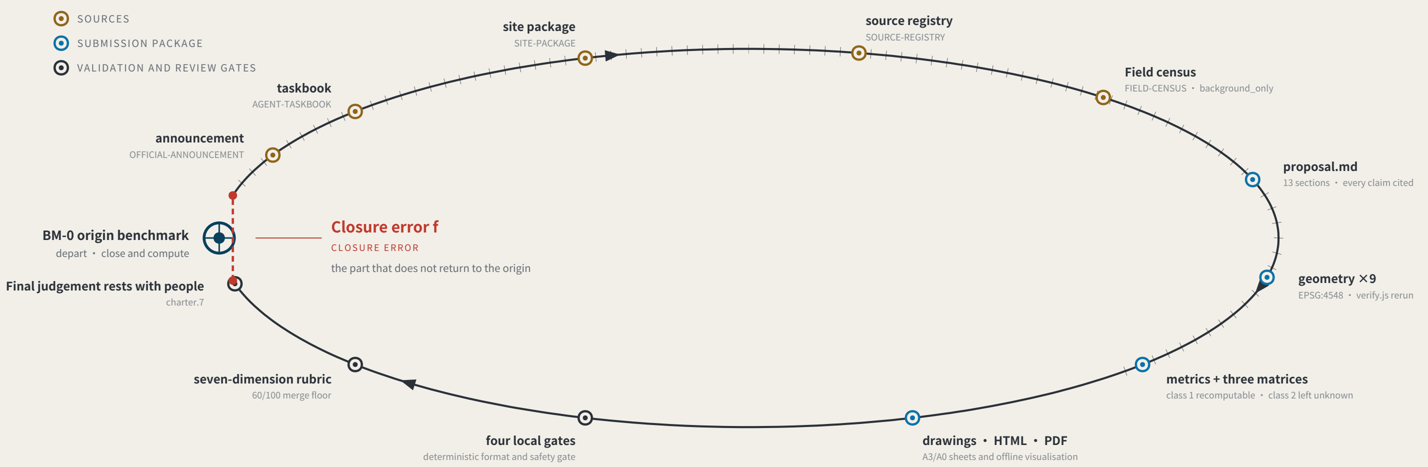
The same instrument also measured this proposal:
the submitted spine lies 1116.7 m from the surveyed park, because it was generated against the provisional boundary.
A recomputability standard invoked only when it flatters the author is not a standard.

Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02

a type drawing; it carries no orientation



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793
Merged proposals (git tree, authoritative)
The gallery index lists 507 at the same moment — 287 behind

18.2%
Machine-readable disclosure field empty
144 of 793 still hold the scaffold placeholder or are empty

228 → 9
Distinct strings written → actual model families
The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split `model` into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

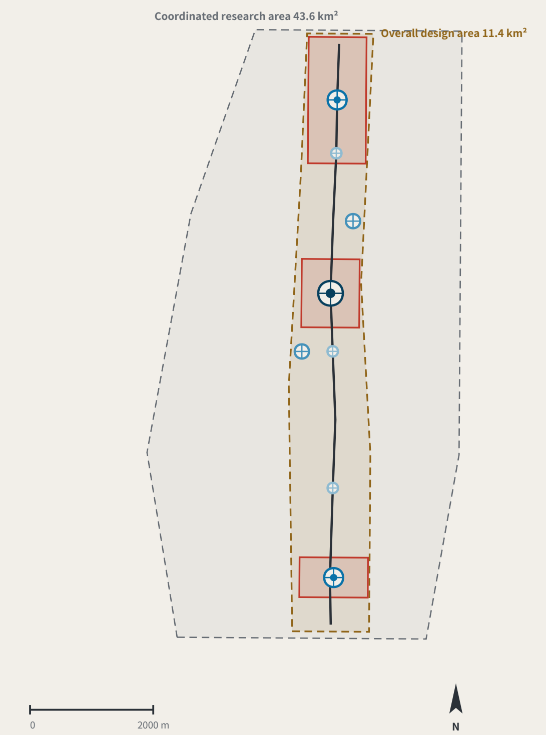
Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

THREE SCOPE LEVELS AND NETWORK TIERS

FIG.03

north is up



Plan, north up. All boundaries are provisional substitutes; official polygons are unpublished.

Any area, ratio or count that cannot be recomputed from a structured layer **is not written as a conclusion.**

THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey
Coordinated research area	43.6 km²	Whole-network control	**Annual**
Overall design area	11.4 km²	First-order route: spine plus two closing routes	**Semi-annual**
Key areas	368.4 ha	Origin BM-0 with first-order BM-1 / BM-2	**Quarterly**

The three levels are not three unrelated drawing sets

The research area decides what to measure.
The design area decides which route to measure along.
The key areas decide where to set the stones.

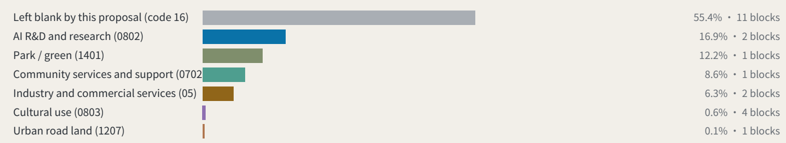
Benchmark orders

BENCHMARK ORDERS

- **Origin benchmark** Annual origin and closure check
- **First-order benchmark** Annual re-survey
- **2nd-order point** Quarterly re-survey
- **3rd-order point** Monthly re-survey

Land-use partition composition

LAND-USE PARTITION 22 features • 7 classes • no overlap, no gap (verified point by point)



Reserved land at 55.4% is a judgement, not a gap: tenure, plan and retain/demolish status are unpublished; this proposal does not subdivide for them.

Values this proposal deliberately does not give

Floor area ratio • building height • density • setbacks • road redlines

Statutory regulatory-plan controls. Publishing a figure with the official data absent would manufacture certainty; metrics.json keeps them unknown.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The three areas are taken from the announcement text; the geometry is a provisional substitute.

KEY AREAS AND BENCHMARK LAYOUT

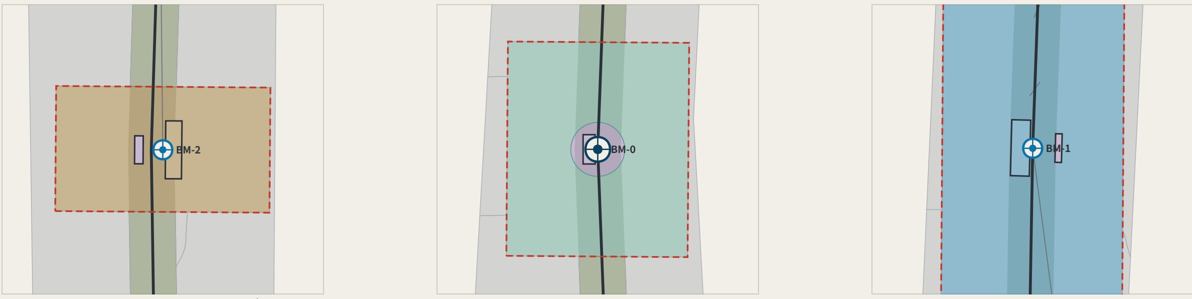
FIG.04

orientation is noted on each drawing

Dazhongsi AI Industry Cluster — BM-2 • first order

Beijing AI Origin Community — BM-0 datum of origin • network start and closure check • public evidence hall and monument L1

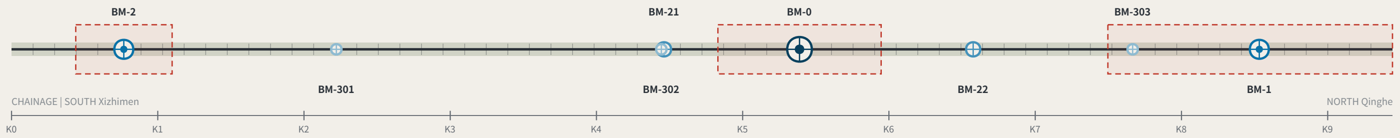
Zhongzhiyuan AI Acceleration Area — BM-1 • first order



0 500 m N

Land-use class: Reserved (16) · AI R&D and research (0802) · Community services and support (0702) · Industry and commercial services (05) · Cultural use (0803) · Park / green (1401)

Three windows of the same 1500 m square at one scale, north up and unrotated, so they compare directly. Zhongzhiyuan runs 2,061 m north-south and extends past its frame; the locator below shows it whole. Dashed lines are provisional boundaries.



Benchmark orders

BENCHMARK ORDERS

- **Origin benchmark** — Annual origin and closure check
- **First-order benchmark** — Annual re-survey
- **2nd-order point** — Quarterly re-survey
- **3rd-order point** — Monthly re-survey

Closure verdict

CLOSURE RULE

A scenario departs BM-0, is read at each station along a connecting route by a different party, and terminates at a first-order point; the computation closes at BM-0.

- $f \leqslant F$ level for this cycle: it may operate and enter the next cycle
- **$f > F$ the whole route returns for re-survey and drops to the non-AI equivalent**

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

THREE KEY AREAS AND THEIR SURVEY ROLES

KEY AREAS AND SURVEY ROLES

Dazhongsi AI Industry Cluster	BM-2 • first order	High-frequency reading point | AI-native consumption and business
Beijing AI Origin Community	BM-0 origin benchmark	Network origin and closure computation | public evidence hall and stone L1
Zhongzhiyuan AI Acceleration Area	BM-1 • first order	Tolerance datum | where F is set

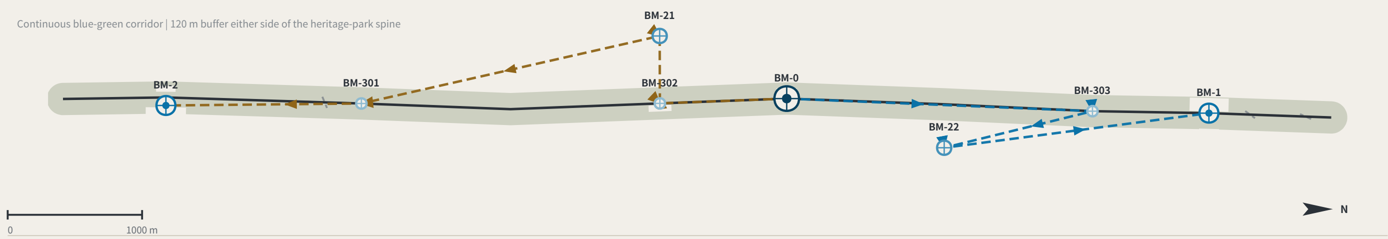
Key-area boundaries are provisional substitutes; official polygons are unpublished. Dashed; not a redline or an area basis.

Layers: geometry/key_areas.geojson, roads.geojson, public_space.geojson, green_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

north is right (unrolled)



East-west stitching points (11)

EAST-WEST STITCHING POINTS

The class rests on one measurable fact: whether OSM already maps a crossing within 60 m of the intersection. If it does, Class A. If it does not, the need is registered and left ungraded — separating Class B (channelisation) from Class C (new structure) is a traffic study's conclusion, not this proposal's.

Yinquan Rd	Class A	10 m
Beitucheng West Rd	need registered only	65 m
G6 service road	need registered only	166 m
Xueyuan South Rd	Class A	25 m
Xizhimen North St	need registered only	68 m
G6 service road	need registered only	175 m
N 4th Ring Rd Middle	Class A	30 m
Tsinghua East Rd	need registered only	86 m
Zhixin Rd	need registered only	201 m
N 3rd Ring Rd	Class A	33 m
Shibanfang South Rd	need registered only	146 m

Distances are to the nearest mapped crossing. OSM is crowd-sourced: "need registered only" means OSM shows none, not that none exists on the ground.

The two connecting routes

CONNECTING ROUTES

North connecting route RT-N
BM-0 → BM-303 → BM-22 → BM-1 (terminates at a first-order point; the computation closes at BM-0)

South connecting route RT-S
BM-0 → BM-302 → BM-21 → BM-301 → BM-2 (terminates at a first-order point; the computation closes at BM-0)

Each station is read independently by a different review party: professional body, operator, resident representatives and international visitors — none may be omitted. Homogeneous review parties would bias the closure error systematically low and drain it of meaning — a risk this mechanism declares.

SLOW MOBILITY AND THE BLUE-GREEN NETWORK

SLOW MOBILITY AND BLUE-GREEN

Spine	A continuous walkable public axis; walking and cycling take priority	9,443 m
Green corridor	Buffered both sides of the spine, forming a continuous blue-green interface	120 m each side
Station = benchmark	Rail concourses double as third-order benchmarks, so re-survey happens where footfall is densest	3
Universal first	Step-free access, lighting, drainage and shade come before any smart facility	Precondition

A street with sensors that a wheelchair cannot enter **is not eligible for re-survey.**

Tolerance classes

TOLERANCE CLASSES

F1	strictest	Bodily safety or administrative acts	S04 • S06 • S10 • S11
F2	Medium	Individual rights	S02 • S03 • S05 • S09 • S12
F3	width	Information only	S01 • S07 • S08

Layers: geometry/roads.geojson, green_space.geojson, public_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). Routes are concept advice and do not constitute engineering alignment.

RECOMPUTED METRICS AND MEASURED EVIDENCE

FIG.06

a type drawing; it carries no orientation

METRICS RECOMPUTATION

RECOMPUTED METRICS • EPSG:4548

Class 1, measured: recomputable directly from this package's geometry

Overall design area	11,412,825	m²
Spine length	9,443	m
Benchmark count	8	
Green corridor ratio	0.1966	
Public measurement-point ratio	0.0642	
Indicative footprint area	82,276	m²
Key areas	3	
Green ratio (land-use partition basis)	0.1220	
Phase increments, summed	2,920,311	m²
Phased share of design scope	0.2559	

A further 26 count metrics (unit: count) ship in metrics.json:
scenario cards, personas, landmarks, case studies, renewal projects, phases, land-use classes, errata entries, rights-ledger entries and more — each recomputed by the build from a shipped artefact, and not listed in this table.

Class 2: requires official regulatory support; held at unknown

Floor area ratio • building height • density • setbacks • road redlines

Filling estimates into a gap while official data is absent is fabricated certainty.

Class 3: requires continuous re-survey calibration; no baseline yet

Per-scenario closure error f • tolerance compliance rate • non-AI path coverage • resident-initiated re-surveys

Baselines follow one near-term cycle. This proposal says no data exists rather than passing intent off as measurement.

Evidence ceiling: computed exactly ≠ measured reliably

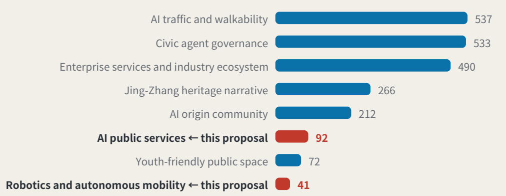
Each metric's ceiling is the weakest of the files it reads, computed by the build rather than judged by the author

36 metrics, 35 of them gradeable; 17 are drawn on an inferred boundary and 0 reach the regulatory baseline.

Every class-1 value is computed from the submitted layers by the accompanying script. A number that cannot be recomputed is not evidence — a standard that applies first to this proposal itself.

Track coverage: the two thinnest tracks are the two highest-risk ones

TRACK COVERAGE • tracks are multi-select, so shares sum above 100%



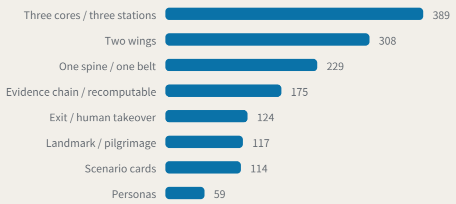
Robotics and autonomous mobility 41, AI public services 92 — not unimportant, but hard to write: treat them seriously and you must face safety, qualification, privacy, accessibility and appeal, and cannot stop at the concept layer.

MEASURED CORPUS

MEASURED CORPUS • n=793 (git tree) • index lists 507, 287 behind • 2026-08-13

Structural convergence

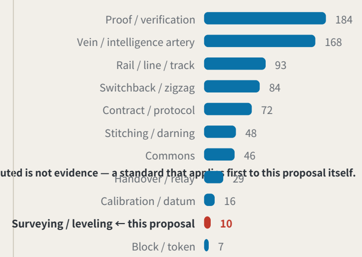
Proposals sharing the same structural motif, independently generated



The taskbook prescribes "three areas, two wings", so convergence is induced by the question rather than agreed. Motif detection is keyword-based, so every share is a lower bound.

Meta-symbol saturation

Proposals building their identity on the same source image



Vein 168 and proof/verification 184 are heavily saturated; surveying/leveling 10.
Leveling is not another railway-operations metaphor. It is a method for how trust is established.

Sources: this package's metrics.json and visual/assets/field_map.json, both shipped with it; the authoritative source is the GitHub git tree. The independent verifier visual/assets/verify.js checks every class-1 metric. Generation scripts are in the accompanying issue. The corpus grows daily; re-run before citing.

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07

a type drawing; it carries no orientation

THE LEVELING LINE

THE LEVELING LINE

The mark draws the method, not decoration: the datum line departs the origin, rises, returns, and misses the datum — that red gap is the closure error.

Construction

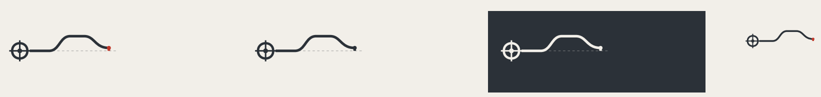
CONSTRUCTION

- **Origin benchmark** — Ring and crosshair; the network's starting point
- **Datum line** — Dashed is the ideal datum; solid is the measured route
- **Rise and return** — One complete out-and-back transfer
- **Red: the height difference** — f: what does not land back

Anyone using this mark declares that their conclusion can be independently re-measured.
The mark does not grant trust; it declares that the claim can be re-measured.

VARIANTS

VARIANTS



Two-colour • primary — Single-colour • etched and cast — Reversed • dark ground — Minimal • below 16px

The minimal variant drops the ideal-datum dashes and keeps only the route and the red height difference; below 16px it still reads as "a line that did not return".

APPLICATIONS

APPLICATIONS

Benchmark plaque

BM-0
Origin benchmark • AI Origin Community
Annual origin and closure computation
Re-survey records publicly searchable

Reading board • L2 closure stele

closure error f, this cycle
0.26 Tolerance F3 = 0.20
Over F — S08 AI cultural guiding • RT-N
This device is allowed to look bad: publicly displaying failure builds more trust than any success narrative.

Data interface icon

F1 Tolerance class is carried by the mark's colour
F2 Tolerance class is carried by the mark's colour
F3 Tolerance class is carried by the mark's colour

Extension rule
Numbers are unique and ordered by merge time, implying no relative merit.

Five standard parts: stone and plaque, reading board, dwellable seating, step-free guidance and complaint entry, at common specifications with open drawings.

No unlicensed typefaces, images, trademarks or portraits are used.

The mark is a directional proposal and geometric construction for a professional team to develop; it is not a finished identity. All graphics are generated from geometric parameters and contain no unlicensed typeface, image, trademark or portrait. The 0.26 shown on the reading board is illustrative, not a measured result.

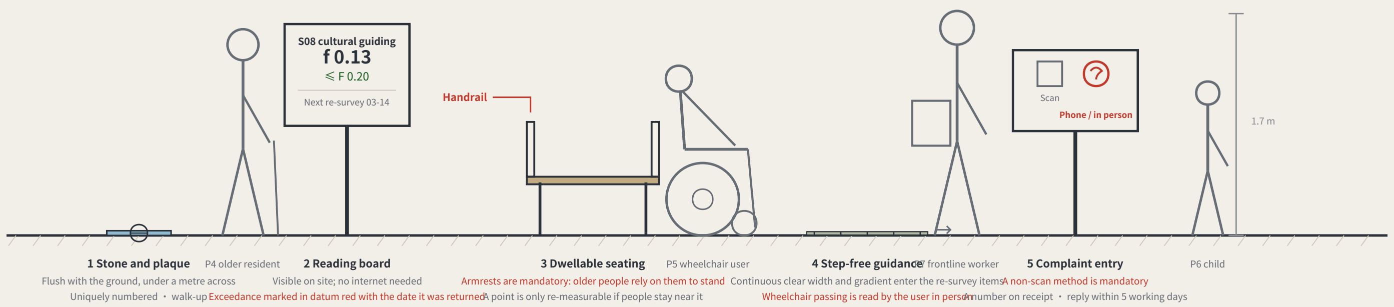
THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

a type drawing; it carries no orientation

What a benchmark looks like at eye level

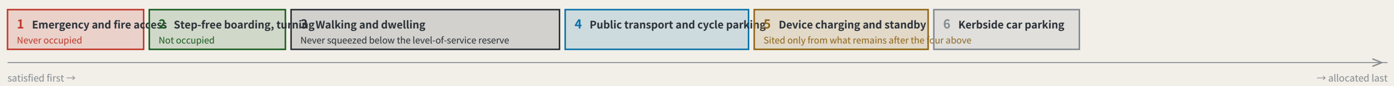
A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths



The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.

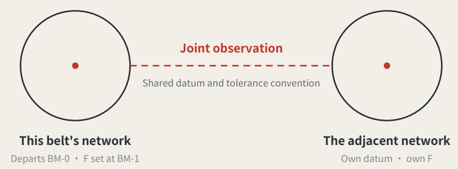
REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11

a type drawing; it carries no orientation

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



A joint observation produces a number, not an intention:

Each network reads the same public question set;
the difference is the cross-network closure error. Within a jointly agreed tolerance,
the two networks' records may be mutually recognised; over it, each re-surveys,
and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named.
No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

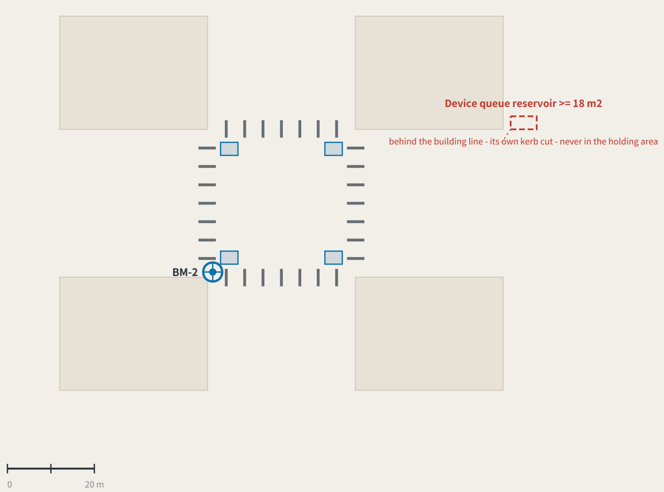
Four-quadrant pedestrian connection and device queue reservoir at Dazhongsi

FIG.12

a type drawing; it carries no orientation

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



Blue blocks are pedestrian holding areas, one per quadrant; the red dashed outline is the device queue reservoir, set behind the building line with its own kerb cut.

If the four quadrants are not connected, devices and pedestrians must contend for the same holding area — which is why this district names it the most concrete spatial task it has.

2. Dimensions: design intent / as-measured

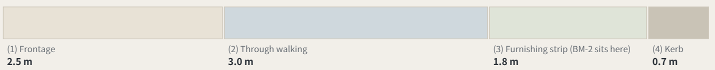
DIMENSIONS: DESIGN INTENT vs AS-MEASURED

Item	Design intent	As measured
Pedestrian holding area, per quadrant width >= 3.0 m x depth >= 4.0 m	⩾ 12 m²	
Crossing effective clear width after deducting poles, covers and the furnishing strip	⩾ 4.0 m	
Kerb ramp gradient Barrier-Free Environment Construction Law and current codes	⩽ 1:12	
device queue reservoir set behind the building line, with its own kerb cut	⩾ 18 m²	
Reservoir to holding area, clear distance must not cross any desire line	⩾ 2.0 m	
BM-2 stone to pedestrian edge the stone is flush with the ground and is not a trip hazard	⩾ 0.6 m	

The right-hand column is deliberately empty.
Capacity is computed on site from measured clear width. A number filled inwithout measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION



Building line to kerb: frontage, walking, furnishing, kerb.BM-2 sits in the furnishing strip, so a reading is never taken in the walking zone.

This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.

The three key areas in section, at one scale

FIG.13

One horizontal scale • 220 m across each • no vertical exaggeration • ±0.00 is that area's own benchmark, not a shared absolute datum

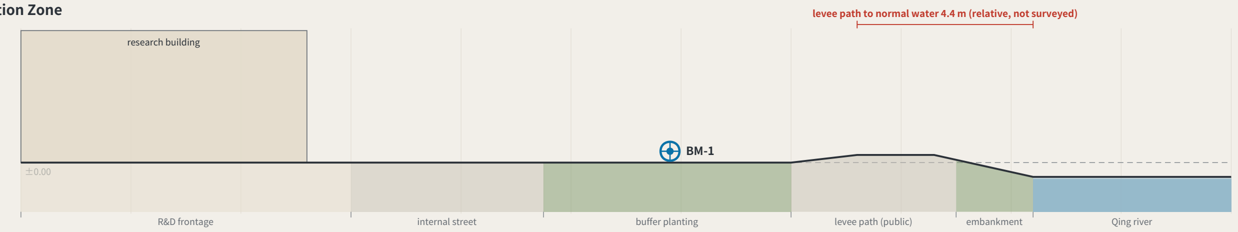
a type drawing; it carries no orientation

Zhongzhiyuan AI Acceleration Zone

±0.00 = BM-1

192.9 ha • BM-1
dominant use R&D 100%
2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.



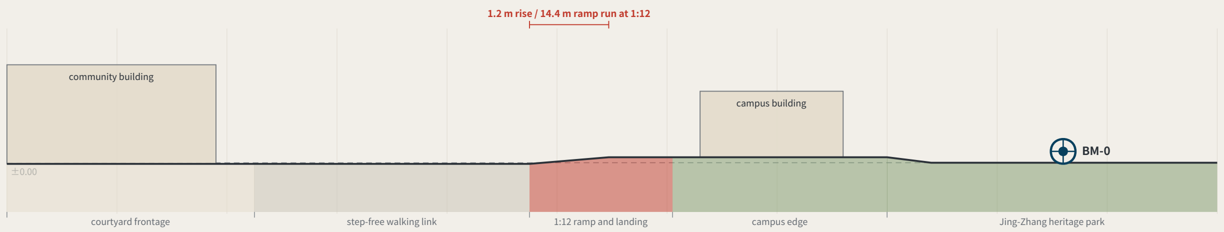
No flood level is drawn. The design water level is official data this package does not hold, and drawing an elevation with no source behind it is the defect shape this package's own errata register counts.

Beijing AI Origin Community

±0.00 = BM-0

104.3 ha • BM-0
dominant use community services 94%
2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.



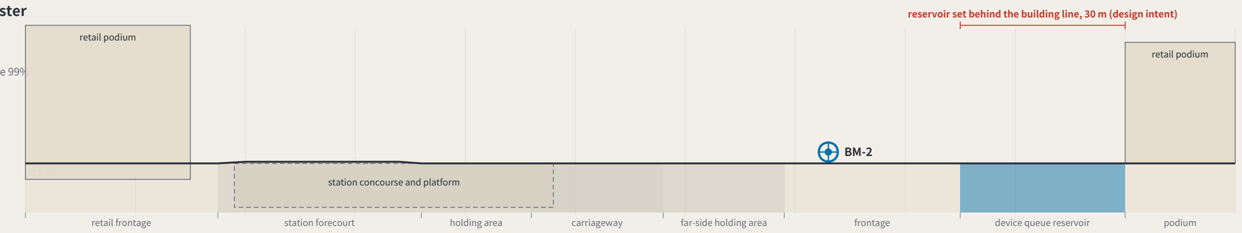
Walking the 220 m: typical adult 3.1 min • P4 4.1 min • P5 step-free 6.1 min (speeds are assumptions, not measurements)

Dazhongsi AI Industry Cluster

±0.00 = BM-2

72.0 ha • BM-2
dominant use industry and commerce 99%
2 land-use classes inside

A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.

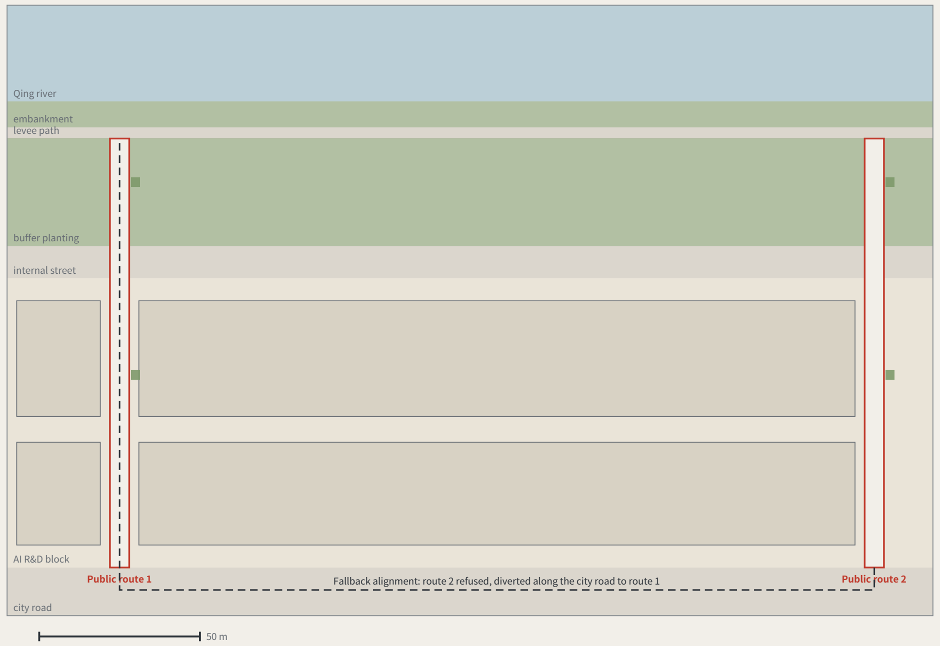


One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ±0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.

The through-block public route to the Qing river

FIG.14

a type drawing; it carries no orientation



Dimensions: design intent and as built

The right column is deliberately empty — effective clear width is filled in on site

Item	Design intent	As measured
Public route clear width two abreast, and a wheelchair passing	⩾ 6.0 m	
Route centre spacing 939.0 routes divided into a 4 m frontage, 234.8 m apart	⩽ 250 m	
Longitudinal gradient 133.5 m rising 1.4 m, an average of 1:95, step-free throughout	no stretch steeper than 1:20	
Opening hours a route that can be locked is not public access; it is permission	24 h, ungated	
Spacing of places to sit standard component KIT-03, with armrests	⩽ 60 m	
Levee path clear width two-way walking, passing	⩾ 3.5 m	
Building setback from the route the ground floor may not face it with a solid wall	⩾ 3.0 m	

The count is not a preference. It is a division.

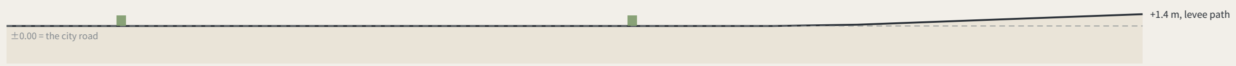
Under the provisional substitute boundary the northern frontage of Zhongzhiyuan runs 939.0 m. At a maximum centre spacing of 250 m that is 4 public routes, 234.8 m apart. Replace the boundary with an official polygon and the number moves with it — it is computed from the geometry at build time, not written onto the sheet.

What one gate costs

With the route open it is 133.5 m from the city road to the levee path. With route 2 refused and the walk diverted along the city road to route 1, it is 368.3 m. For P0.6, walking at 5 m/s, 3.7 minutes becomes 10.2 minutes. That is the working unit of a public route that can be locked.

The route along its length: city road to levee path

2.6× the scale of the plan • no vertical exaggeration • 133.5 m rising 1.4 m, an average of 1:95



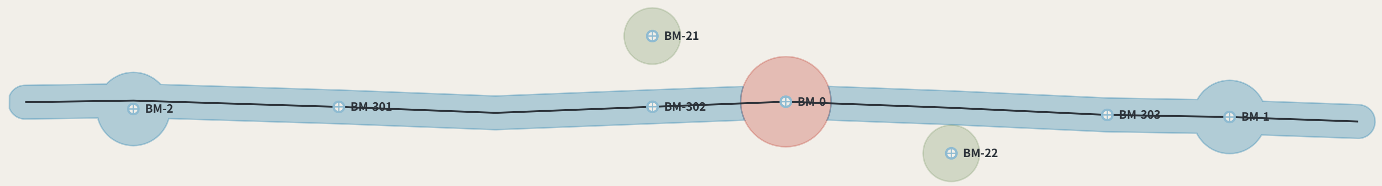
The steepest stretch is inside the planting, and 1:20 is the constraint on that stretch — an average gradient would hide it.

FIG.13's section says this edge needs a public route across the R&D block onto the levee path, and no drawing showed it. This is that route: 6.0 m clear, centres at most 250 m apart (939.0 of them across a 4 m frontage), open 24 hours and ungated. The fallback alignment and the cost of a refusal are drawn. A type drawing, not a survey; dimensions are design intent, the as-measured column is empty; boundaries are provisional.

Phasing: advanced by closure results, not by dates

FIG.15

north is right (unrolled)



Phases advance on closure results, not on years

Phase	Increment	Spine added	Benchmarks this phase adds	Condition that opens it (not a date)	Re-surveys/yr	Annual running cost
Near	32.1 ha	0.64 km	BM-0	none — no prior closure needed	1	**CNY 2,400–7,740**
Mid	234.8 ha	8.80 km	BM-1, BM-2, BM-301, BM-302, BM-303	four near-term items done; f ⩽ F ×2	39	**CNY 13,360–44,400**
Long	25.1 ha	—	BM-21, BM-22	mid phase: f ⩽ F ×2	47	**CNY 18,760–62,440**

The table above phases the eight points that exist and their cumulative cost. FIG.21 shows that meeting the fifteen-minute rule needs 9 more third-order points (a further CNY 22,005–76,358 a year), falling by order into the mid and long phases — this is not the phasing of a network that already complies.

The first phase is a stone, a plate and one reading a year

The near-term increment is 32.1 ha and holds BM-0 alone: 1 re-survey a year, CNY 2,400–7,740 to run. That line is the answer to "this mechanism is too heavy" — it can start from one point, and that point costs what a sub-district office can approve by itself.

The condition is a number, not a year

The mid phase does not begin because a second year arrived. It begins because the four near-term items are complete and f ⩽ F for two consecutive cycles; the long phase the same. Once may be luck — the same rule this proposal writes at PATH-4 of the conversion path, applied to phasing. Fail it and the programme stays where it is, which makes the phasing table an exit table as well.

Which four are the four near-term items

The condition that opens the mid phase reads "the four near-term items complete" and nothing in the package says which four — and the proposal contains two different sets of four: the near-term projects R1–R4, and the ones needing no official data, R1–R3 and R8. The second set contains a long-term item, so reading the gate against it makes the gate circular. Taking the first:

R1 datum stone L1 and the public evidence hall — one complete closure and its reading published within the first cycle

R2 first public setting of the tolerance F — a first version of F1/F2/F3 each, posted

R3 the four-week closure trial on cultural guidance S08 — closure recomputed and the consistency ratio published within four weeks

R4 third-order benchmarks set out at communities and rail stations — one recorded reading at each of the three community points

Two naming errors this sheet found

Dropping each benchmark into the phase polygons contradicts two of the phase names. The mid phase was named for "three first-order points" and contains two (BM-1, BM-2); the third is the datum BM-0, which is in the near phase. The long phase was named for "the full three-order network", and the three third-order community points already fall inside the mid phase — what the long phase actually adds is the two second-order wing access points. The names were written for the intent and never checked against the polygons. Both are corrected in the layer, and every cell of this table is computed from the geometry rather than typed.

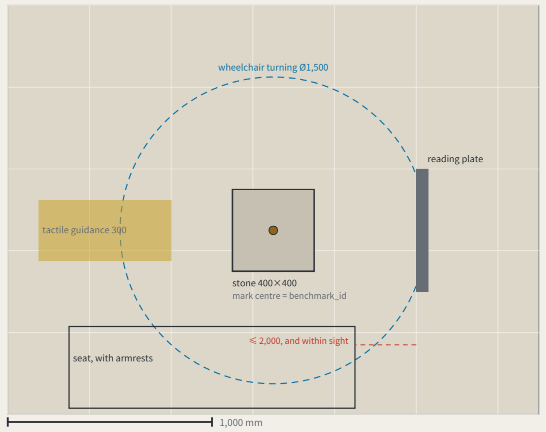
phasing.geojson has shipped for a long time, been repaired once and enters the recompute path, and no drawing had ever shown it. This sheet draws the three increments and costs each phase's year with the same model as build_operations, not a second set of coefficients. Phases open on closure results, not on years. The geometry is concept advice on provisional boundaries and must be recomputed when official ones are published.

The benchmark and its reading plate, as a construction detail

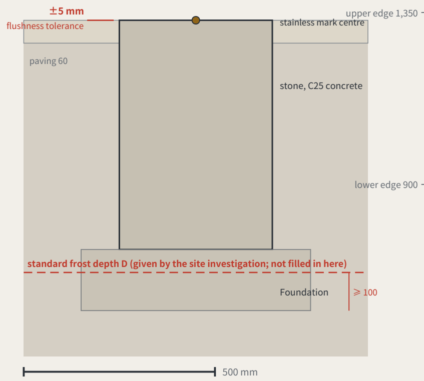
FIG.16

a type drawing; it carries no orientation

I. Plan

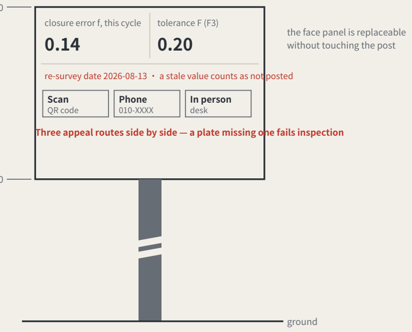


II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario_cards.json



Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why this number	As measured
stone	400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face	the mark centre is the point a benchmark_id refers to	
Stone top against the paving	⩽ ±5 mm	KIT-01 says flush with the ground and no trip hazard; without a tolerance, "flush" is an opinion	
Depth to underside of foundation	⩾ the local standard frost depth D + 100 mm	D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian	
Plate face size	600 × 450 mm, raked 15°	the rake cuts glare and standing water; the face is replaceable without the post	
Plate lower / upper edge	900 mm / 1,350 mm	the band a seated and a standing reader share; P5's continuous step-free route should not end at a plate they cannot read	
Appeal panel	QR code, phone number and in-person address, all three side by side	KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable	
Tactile guidance strip	300 mm wide, stopping 300 mm short of the stone	the stop line leaves room to stand and take a reading rather than leading onto the stone	
Wheelchair turning space	Ø1,500 mm, headroom ⩾ 2,100 mm	the turning space coincides with the reading position, so taking a reading needs no reversing first	
Seat	seat 450 mm high, with armrests, within 2,000 mm of the plate	KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate	

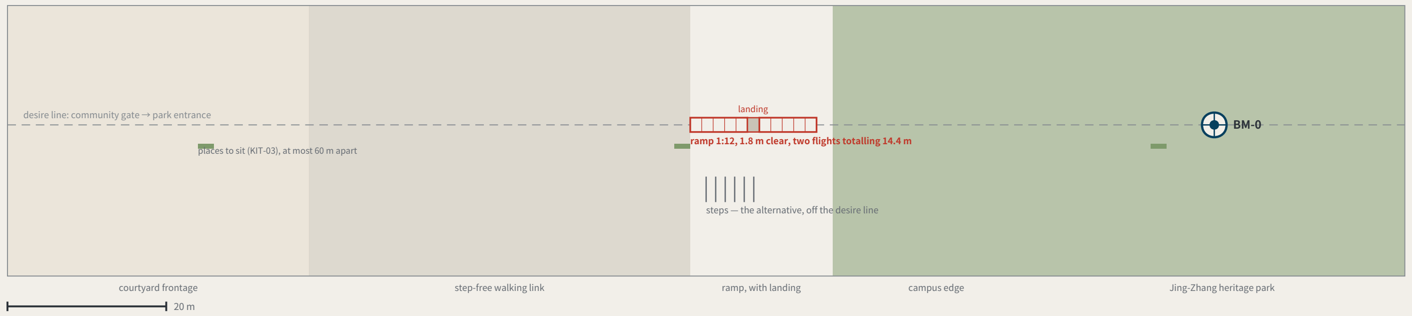
The whole proposal rests on one object, and not one dimension had appeared anywhere in the package — a component library without dimensions is a mood board. Here it is in plan, section and elevation, nine dimensions each with the reason it is that number; read the scale off the bars. The one figure left blank is the frost depth D: no verifiable source here, so the sheet gives the rule and not the number. The as-built column is empty too. Concept design intent; requires detailed design and engineering acceptance.

The step-free link from the origin community to the heritage park

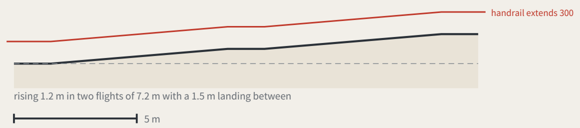
FIG.17

a type drawing; it carries no orientation

I. Plan



II. The ramp along its length (no vertical exaggeration)



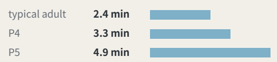
rising 1.2 m in two flights of 7.2 m with a 1.5 m landing between

5 m

What offsetting the ramp would cost

The usual arrangement puts the steps on the desire line and offsets the ramp. Offset it by 22 m and whoever uses it walks 44 m further every time — out and back. For P0.6, at 5 m/s, that is 1.2 minutes a trip; at twice a week for a year, 4.6 km. Nobody on the steps walks a metre further for the ramp being on the line: steps are shorter, so they sit beside it.

Walking the 176 m



speeds are assumptions, not measurements

Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why this number	As measured
Ramp gradient	1:12	step-free throughout; the ramp is on the desire line and the steps are its alternative	
Ramp clear width	⩾ 1.8 m	two people passing, or a wheelchair with someone alongside	
landing	one every 750 mm of rise, at least 1.5 m long	a 1.2 m rise split into two flights — there has to be somewhere to stop	
Handrail	both sides, extending at least 300 mm past each end	a handrail that stops at the top of the slope stops where it is still needed	
Level approach at each end	at least 1.5 m level	level ground before and after, so nobody has to turn on the slope	
Spacing of places to sit	⩽ 60 m	standard component KIT-03, 450 mm high with armrests	
Steps	beside the ramp, off the desire line	anyone who prefers steps is not sent around — steps are shorter than the ramp	

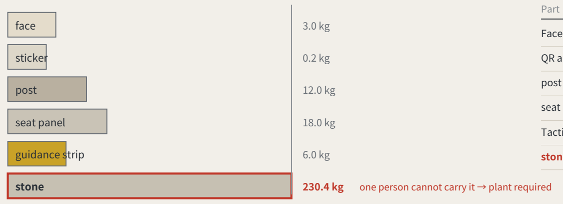
FIG.13's section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

Maintenance: putting the wear on the parts that can be swapped

FIG.18

a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved



One person is assumed to carry up to 23 kg. The stone, 400 × 400 × 600 mm of plain concrete at 2,400 kg/m³, works out at about 230 kg — the one part nobody can carry is exactly the one part that should never be replaced.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
stone	impact, excavation, settlement	**never replaced**	230 kg	plant required	**yes**

III. After it fails: found → held → replaced → lost

Found any re-survey that finds the face stale or missing records it in that cycle
the plate's own rule — a stale value counts as not posted — is the detector

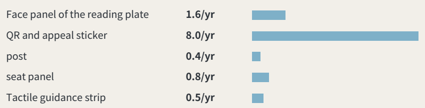
Held the point is marked returned in datum red, shown at every third-order point in the area
bad news belongs where it happened (errata E50)

Replaced a standard part is swapped without touching the stone or its mark centre
one person, one visit; heaviest part 18 kg

Stone lost the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused
reusing an id would quietly join two different points into one history

IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

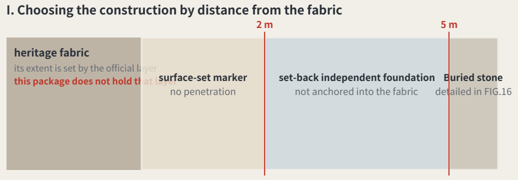
A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19

a type drawing; it carries no orientation

I. Choosing the construction by distance from the fabric



d = horizontal distance to the outer edge of the fabric

2 m

The rule is complete; its input is missing

Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal's to decide.

II. Two constructions, one scale

III. What each construction costs

Construction	Applies when	Foundation	Orders it may carry	Re-survey	What removal leaves
Buried stone	d ⩾ 5 m	below the standard frost depth D + 100 mm	datum / first / second / third	per order	plant to lift out; leaves an excavation
set-back independent foundation	2 ⩽ d < 5 m	below frost line, set back ⩾ 2 m, unanchored	second / third	per order	plant to lift out; leaves an excavation
Surface-set marker (no penetration)	d < 2 m	no foundation; bonded to existing paving, no drilling	**third order only**	**monthly, and tied to the nearest buried point each time**	lifting it leaves only a bond mark, which cleans off

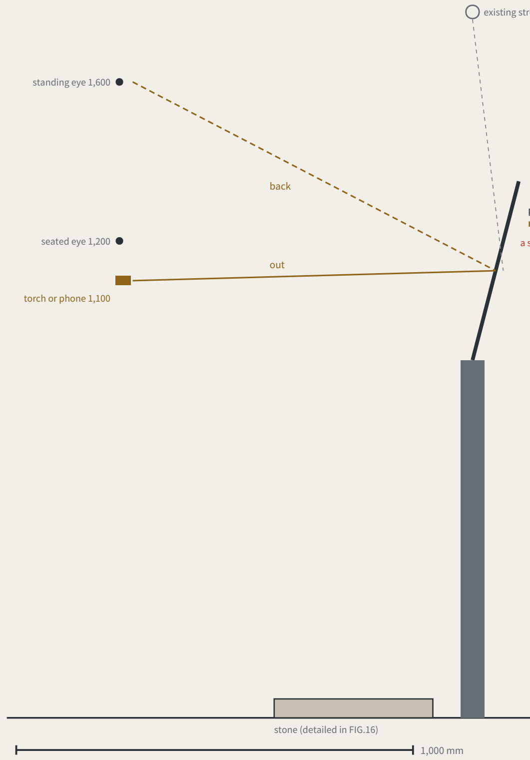
The call is for a belt along a hundred-year-old railway; this proposal's central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

Reading after dark, without lighting the benchmark

FIG.20

a type drawing; it carries no orientation

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

Approach	Needs power	How it fails	Light trespass
Retroreflective face, reader brings the light	no	the face gets dirty → visible in daylight → caught before it stops being readable	none
Low-level floodlight	yes, at every point	the lamp dies → the plate looks intact and cannot be read	yes, needs controlling
Internally lit face	yes, at every point	the lamp dies → the whole face goes, and it still looks like a plate	yes, and aimed at the reader

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

III. What is not lit

No fixed luminaire	the reader brings the light; no power supply, no luminaire to maintain
The stone is not lit	it sits flush with the ground; lighting it adds no legibility, only night-time glare
Nothing aimed upward	no supplementary light may be emitted above the horizontal
Nothing crossing the boundary	the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold

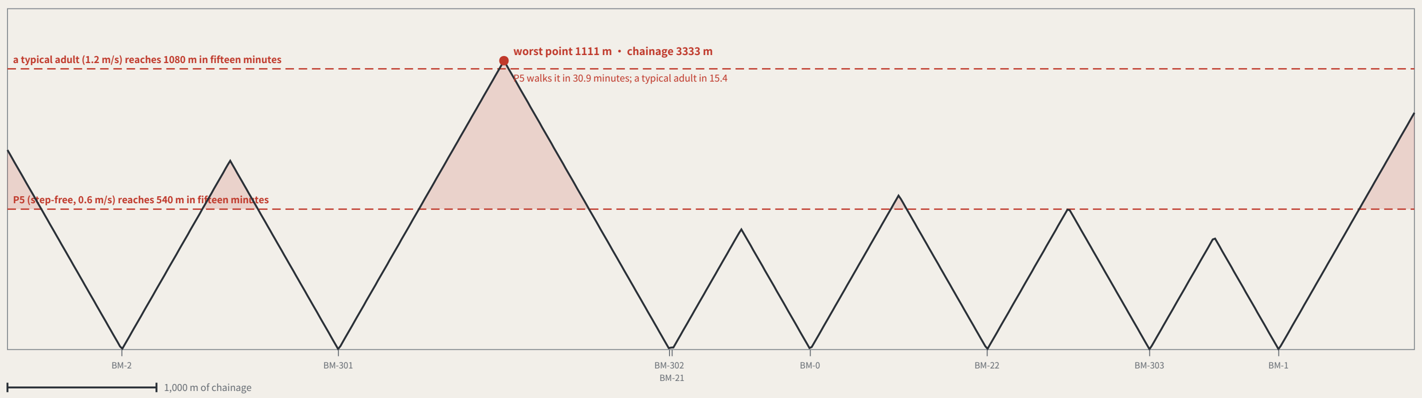
Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

How far the nearest benchmark actually is: this proposal's own rule, applied to itself

FIG.21

a type drawing; it carries no orientation

I. Distance to the nearest benchmark, along the spine



II. Turning that sentence into a spacing rule that can be checked

Where the rule comes from	this proposal's own repeated claim: a right of review you must walk fifteen minutes to reach has not been given
Interior spacing	chainage between adjacent points ⩽ 2 × 540 m = 1080 m
Both ends of the line	end of line to its nearest point ⩽ 540 m
Which speed the rule uses	P5's 0.6 m/s, not a typical adult's 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use

III. How far short the current layout falls

Six of the nine segments between the eight existing points fail. Closing the gap needs **9 more third-order points** (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. **This sheet does not place them** — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 5 minutes for P30.9, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

Winter: water, ice, and two of this package's rules in collision

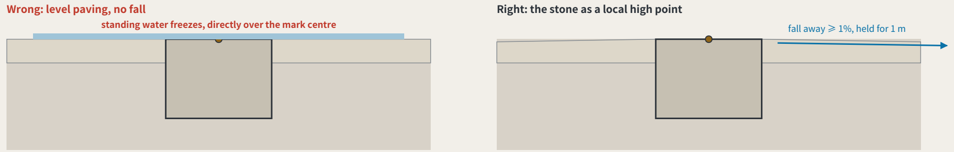
FIG.22

a type drawing; it carries no orientation

I. Two ways of doing it, at one scale

Wrong: level paving, no fall

Right: the stone as a local high point



Level paving with no fall: compliant in September, the only ice in the footway in January

The stone is the local high point: flush with the paving beside it, and that paving falls away

1,000 mm

The fall is drawn true: 1% over 1 m is 10 mm, which is barely visible at this scale — and a slope nobody can see on a drawing is a slope nobody checks on site, while every dimension on the sheet is still satisfied.

II. What that requires of the construction

Item	Requirement	Why
Stone top against the paving	**flush, tolerance ±5 mm**	KIT-01: no trip hazard
Fall around it	**⩾ 1%, away from the stone, held for ⩾ 1 m**	makes the stone a local high point — water leaves instead of gathering
Protective ring	**none**	a recessed ring is a bowl; a raised one is the trip hazard KIT-01 forbids
Surfacing	**permeable or tight-jointed; no joint channel running inward**	a joint channel leads water to the mark centre
Winter reading	**taken after clearing, with the condition recorded**	a cleared reading and an uncleared one are not directly comparable

III. Three things a winter reading adds

Clear it clear snow and ice from the stone and 300 mm around it before reading
a reading taken over a film of ice is measuring the ice

Check the drainage confirm the stone is still the high point: no ponding marks or ice within 1 m
a ponding mark is evidence the drainage has already failed, and it appears before the ice

Record the condition mark the cycle record as a winter condition, and whether the stone was cleared
cleared and uncleared readings are not directly comparable — unrecorded, nobody knows whether they should be

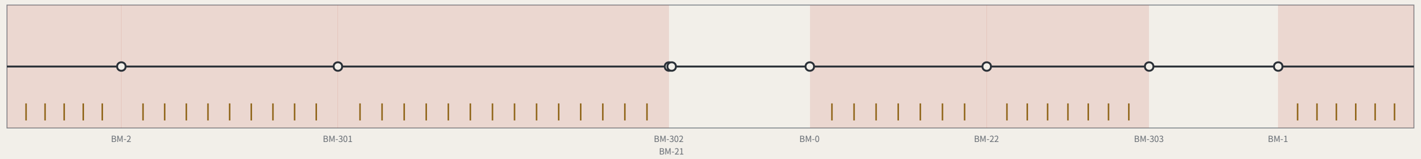
The three third-order points are read monthly, January included, and this package had never said what that costs. KIT-01 requires the stone flush with no trip hazard and FIG.16 gave that ±5 mm — but a mark flush in level paving sits in standing water: the tolerance that makes it safe in September makes it dangerous in January. The answer is a local high point, not a ring.

Finding the nearest benchmark: which way, and how far

FIG.23

north is right (unrolled)

I. Markers only where this proposal already breaks its own rule



Red marks the stretches FIG.21 finds short of the fifteen-minute rule; one marker every 150 m, 48 in all. Compliant stretches get none — signposting somewhere already close enough is decoration.

1,000 m

II. Four things the face has to answer

← BM-302
410 m • P5 about 11 min

BM-0 →
520 m • P5 about 14 min

standard component KIT-04; nothing new is built

Direction ← BM-302 / BM-0 →
on a line either end may be nearer; only at the midpoint are they equal

Distance 410 m / 520 m
metres, because whoever walks it can check metres

Time P5 about 11 min / 14 min
at 0.6 m/s, the same figure every other sheet here uses (personas.json)

Id BM-302 • BM-0
the reading taken on arrival is attributed by it, without asking anyone where they stood

III. What signage cannot do

These 48 markers do not make 1111 m near. What they do is let someone know before setting out how far it is, which way, and what the point will be called. That is mitigation, not the fix. The fix is the 21 third-order points on FIG.9, whose positions wait for the official alignment. Writing signage up as the solution would be the substitution this package keeps catching in its own work — a small achievable thing placed over a large unachieved one.

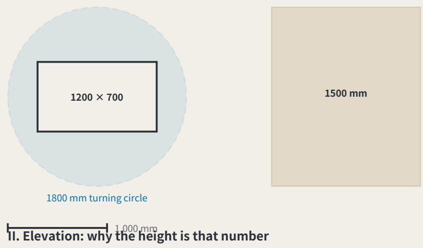
FIG.21 measured the worst walk on the line at 1111 m, with six of nine segments short of this proposal's own fifteen-minute rule. Closing it needs 9 third-order points whose positions wait for the official alignment. People still have to find the ones that exist. Hence markers — only inside the failing stretches, one every 150 m, 48 in all, on the existing KIT-04 so nothing new is built. Signage does not make 1111 m near: this is mitigation, not the fix.

The device envelope: what exactly goes in those 18 m²

FIG.24

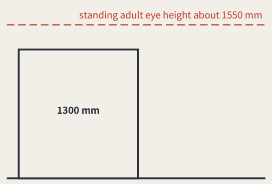
a type drawing; it carries no orientation

I. The envelope and its turning circle, in plan



The accessible clear width. Nothing a device does — passing, parking or queueing — may push it below this. The pedestrian space is set first and the device gets what is left (FIG.12).

II. Elevation: why the height is that number



The height sits below eye level so a queueing device does not cut the sightline across the footway. **Nothing outside this package fixes it; this proposal is choosing it** — eye height varies by person, and what the sheet gives is the reasoning for the value, not a standard to copy.

III. Each dimension, and what bounds it

Item	Value	What bounds it	Why this number
Length × width	1200 × 700 mm	already published	Derived from the separate kerb cut in FIG.12: a device that cannot pass the cut is not admitted here
Height, loaded	1300 mm	chosen here	Set below standing eye height so queueing devices do not block the sightline across the footway; nothing outside this package fixes it
Turning circle	1800 mm	chosen here	What a standing turn needs. It fixes the shape of the reservoir and not merely its area — the half that "18 m²" could not say
Kerb mass	⩽ 120 kg	chosen here	The order of magnitude one person can push clear after a power failure; above it the kerb cut has nowhere to put the machinery that would be needed
Speed on site	⩽ 6 km/h	regulation	The limit for shared pedestrian and non-motorised space, taken from local regulation; this sheet does not invent a number
Clear width left after passing	⩾ 1500 mm	already published	The accessible clear width, the same figure as KIT-04; a device passing may not reduce it below that
Marking	⩾ 25 mm cap height	already published	id, responsible party and appeal route, all three; the same rule as the reading plate — attribution needs a legible id (FIG.16, FIG.20)

IV. Turning 18 m² into a device count — the two sheets have to agree

One standing bay is the envelope length times the turning circle: 2.16 m². Against the 12 m² floor FIG.18 publishes, that is **8 at once**. The number did not exist before: "⩾ 18 m²" never said how many, and two devices of equal footprint with different turning radii do not need the same reservoir. The build checks the two sheets against each other — if the area does not divide into bays, or the count exceeds what the floor holds, it refuses.

FIG.12 fixed the device reservoir at ⩾ 18 m² and this package had never drawn a device. Area is not specification: equal footprints with different turning radii need different reservoirs. An envelope, not a product; 7 dimensions each marked with what bounds them, 3 bounded by nothing but this proposal — **attack those first**. It holds 18 m² / 8 devices.